

A bi-level distribution mixture framework for unsupervised driving performance evaluation from naturalistic truck driving data

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ABSTRACT

Driving performance evaluations can contribute to fleet management and lead to safer and more economical driving conditions for manned or driverless fleet vehicles. One approach to driving performance evaluation involves quantitative mapping or categorical labeling of skill levels and categorizing of driving patterns from extraordinarily mild to the most aggressive. This paper presents a big data system for driving performance evaluations of drivers and trips using a probabilistic framework. The proposed framework combines a feature mixture model for scoring driving performance through defined objective comparison criteria and a latent style mixture model for classifying drivers by the main driving styles they exhibit. To demonstrate the effectiveness of the proposed models, we perform both quantitative and qualitative experiments. The results show that the former produces an interpretable and normal scorecard model, while the latter helps build an improved clustering model that represents enhanced driver behavior.

1. Introduction

The self-driving truck is very promising in the field of self-driving. Before transitioning to fully autonomous trucks, it is necessary to study the behavior of truck drivers from two aspects. On the one hand, human driving behavior analysis is crucial for accident prevention and energy management. For example, aggressive drivers usually develop abrupt and unpredictable maneuvering styles, which generally result in higher fuel consumption and tend to provoke dangerous situations (Taubman-Ben-Ari et al., 2004). Identifying drivers with this style and providing targeted improvement feedback can significantly reduce fleet management operation costs. On the other hand, driving behavior analysis is useful in designing driving models, as driving models will become the core parts of the algorithms in future self-driving vehicles (Wang et al., 2014). Therefore, an enhanced understanding of human driving behavior is required.

In this paper, we focus on the driving behavior of truck drivers from logistic fleets. The drivers were driving in an open environment, and their performance was mainly influenced by their driving styles but could still differ trip by trip. Fleet managers are extremely concerned about the potential risks posed by improper driving behavior, as any traffic accident caused by unsafe driving events (Fazekas et al., 2014) may lead to delays in the delivery of goods, damages, and even casualties. The costs resulting from truck accidents are much higher than those involving passenger cars. Given that most logistics

trucks are connected vehicles, the challenge that fleet managers face is determining how to handle the massive vehicle sensory data to give a comprehensive evaluation of each driver's performance.

The evaluation involves not only the recognition of the main style for all members but also the investigation of their real-time performance on trips in progress. With the evaluated results, the manager could obtain feedback about each driver's performance during each delivery trip in time to take follow-up corrective actions. Characterizing the driving style plays an important role in observing whether a driver is capable of a transportation task or developing other intelligent systems (Martinez et al., 2018). As recognized, the same driver could exhibit disparate styles under different environmental conditions, such as a delivery trip to an urban or rural area. Hence, more attention should be paid to drivers' dynamic performances on their driving segments. Moreover, traditional ways of driving performance evaluation involve assigning drivers with finite style labels, yet this makes it difficult for managers to rank drivers or trips for further comparison. Thus, quantitative driving performance analysis is also necessary, as it is impossible to differentiate drivers/trips even when belonging to the same style.

Our research work addresses the three challenges shown in Table 1. First, at the driver level, the importance of features is either ignored or subjectively assigned when using clustering algorithms for grouping drivers according to their driving style. We attempt to derive feature

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Table 1

Research focus of our proposed system.

Driver-level	Consideration of feature weights on unlabeled data for better driver style recognition
Trip-level	Scoring of driving trips with a regression function in an unsupervised manner
Engineering application	A distributed big data system handling massive and streaming driving data

weights on unlabeled data for a better driver style recognition effect. Quantitative evaluation (scoring or ranking) of trip-level driving performance is required as well, but it is challenging to build reasonable regression methods due to the lack of labeled data. The last challenging problem is handling large-scale vehicle sensor data for engineering concerns.

To this end, this research enables solving both problems in one big data system. Specifically, big data tools are first used to help train the models offline. In addition to classifying drivers into discrete classes according to their main style, this system also handles the received streaming data and gives each driver a performance evaluation result at the trip level in near-real time. The main contributions of this study are summarized as follows:

- Using the available large-scale vehicle parameters without labels, we build a probabilistic framework including bidirectional mixtures to combine the original distributions of metric features and the latent distributions of underlying styles. This framework is designed to evaluate driving performance at both the trip and driver levels. This preliminary but highly interpretable framework is of statistical significance.
- We propose a scorecard model for scoring driving trips that integrates a feature-direction distribution mixture. By investigating the distribution and statistics of the trip-level metric features over explainable driving features, the particular distributions of the features are first defined. Then, the corresponding cumulative probability density is intuitively connected with the behavior of drivers.
- We present several methods for learning the vector that captures a driver's main style. The first group of vectorized representations of a driver is learned over the trip vectors. The second group contains inputs that add the entropy weight (Zhu et al., 2020) to the first group to help measure the feature importance. Then, we consider each class of driver style as a Gaussian distribution and introduce weighted clustering to improve the clustering performance.
- We pioneered the distributed implementation of our framework with big data technologies. The big data architecture is able to handle the collected big automotive data and implement our framework for near-real-time driving performance evaluation. This system is designed to assist fleet management companies that require instant analytic feedback concerning every driving trip, as well as a driver's overall style.

The paper is organized as follows: a literature review is presented in Section 2. In Section 3, the considered dataset and the data preprocessing are detailed. The proposed framework, trip scoring and driving style clustering methods are given in Section 4. Section 5 introduces the big data system implementing this framework, and a typical user interface is proposed for visualization. Experimental results on synthetic and real-world datasets are demonstrated in Section 6. Finally, the paper is concluded in Section 7.

2. Related work

Many data-driven machine learning methods have been proposed to analyze driving behaviors since they are suitable for processing large

amounts of data and can consistently derive the threshold values or even develop improved algorithms (Martinez et al., 2018; Lin et al., 2014). Elamrani Abou Elassad et al. (2020) presented a systematic review of machine learning models for driving behavior analysis and demonstrated that machine learning models outperformed other conventional approaches. Depending on the various research purposes, a driving performance evaluation output can be either qualitative (aggressive, normal, and cautious) or quantitative (a continuous index from 0 to 1).

The general tendency of the previous literature (Wang et al., 2010; Quintero M. et al., 2012; Van Ly et al., 2013; Vaitkus et al., 2014; Qi et al., 2019) is to group drivers into classes based on the distribution of selected driving parameters or extracted features. These classes, which are most commonly dangerous versus safe or nonaggressive, neutral and aggressive, need to be defined prior to the classification algorithm design. Recently, a systematic comparison of driving style classifications was conducted through five classifiers (Silva and Naranjo, 2020) to show that supervised machine learning methods achieve promising results on this problem. However, such classification-based approaches typically follow the supervised learning paradigm and heavily depend on the obtained labels. When these algorithms are applied in real-world applications, it is quite challenging to collect data from a large number of drivers and ensure a sufficiently large training set for each driver, and the labor costs of the labeling process from authoritative approval channels can be even higher.

In view of this, unsupervised algorithms have proven their applicability and suitability for driving style recognition. For clustering drivers, the K-means algorithm was commonly adopted by Quintero M. et al. (2012), de Zepeda et al. (2021) and Umberto et al. (2019) for automatically dividing drivers into a predefined number of styles. In contrast to such hard-clustering methods, Miyajima et al. (2007) implemented a Gaussian mixture model (GMM) based on car-following behavior and pedal operation spectral analysis. The model achieves a car-following identification rate of 69%, while the pedal spectral analysis provides a classification rate of 89.6% in a simulation environment and 76.8% in field tests. A similar application of a GMM for pedal operation modeling was also reviewed by Wang and Lukic (2011). GMM models the distributions of driving style as a mixture of Gaussians, which will be adopted in this paper as well. As revealed in these studies, the use of predefined driving events (Bagdadi, 2013; Fazekas et al., 2014) could provide valuable information to drivers, such as the frequencies of sharp acceleration, sharp braking, and sharp turning. In addition, another kind of method was to derive statistical features (Vaitkus et al., 2014) from sensory time series, such as the statistics of speed, rotation and steering angle. Spectral clustering was also adopted by de Zepeda et al. (2021) for dynamic clustering-based driving style identification and profiling in response to the changing surrounding environment. Throughout previous driver clustering literature, however, the importance of features was seldom studied when preserving the original structure of the features. Feature importance has been demonstrated to be effective in mass load prediction (Liu et al., 2021) and modeling of lithium-ion battery manufacturing (Liu et al., 2021) but with supervised learning on labeled examples. When only providing features without labels, the clustering methods should also consider the weights over features to shape better clusters.

Driving style classification through continuous indexing is also necessary in some cases. For instance, the trip-level quantitative evaluation of driving performance could help insurance companies develop more fine-grained types of insurance, such as country travel insurance and commuting insurance. In fleet management, identifying a particular trip with high risk could help managers coach drivers to reduce unexpected accidents or economic losses. As a result, the conventional way to group driving styles into discrete classes has its drawback in ranking drivers or trips belonging to the same category.

To quantify accident risk and provide personalized rates, Augustynowicz (2009) considered acceleration and speed as diagnostic

variables and classified driving style in a range within $(-1,1)$, with -1 , 0 and 1 being mild, normal and aggressive, respectively. However, the ranking variable is simply the average value of normalized variables, whose distribution and importance are neglected. In 2014, Bergasa et al. (2014) presented a DriveSafe app to detect inattentive driving behaviors and give corresponding feedback to drivers; it scored their driving and alerted them when their behaviors were unsafe. The distributions of driving events were considered to assign scores using the cumulative probability density of each feature but with manually set feature weights. Ozgul et al. (2018) proposed to rate driving performance with respect to likelihood given trajectory and traffic flow. The scoring strategy involves simple joint probabilities estimated by the frequency of samples that lie in preclustered driving styles. By providing 10 drivers, 1 being aggressive and the others nonaggressive, their proposed framework obtained significantly lower scores on this labeled set for the aggressive drivers compared to their nonaggressive counterparts. Our work is inspired by their ideas but differs in the feature distribution mixture definition and the objective feature weighting strategy. With GPS data collected from a taxi fleet, Wang et al. proposed a deep learning-based representation method to learn representations of driving behaviors, and then a support vector regressor was trained to score driving performance (Wang et al., 2019). Nevertheless, the semantic meaning of latent features rather than those driving events or statistical quantities was lost because a final output without understandable supporting indicators is not readily accepted by drivers. The Gaussian process was adopted by Liu et al. (2020) and Liu et al. (2019) to describe the way to achieve uncertainty quantification that follows a normal distribution. In our case, we also assume that the trip-level quantization output follows such a distribution. However, manually labeled training data were still required to build regression models.

Last, big data analytics have been reported to possess the ability to guide the creation of safer transportation systems (Neilson et al., 2019). Early in 2014, Johanson et al. (2014) developed a big data framework to explore the opportunities of leveraging big automotive data for knowledge-driven product development. Pevec et al. (2019) showcased the collection, storage, and analysis of the data generated by an electric cleaner fleet by proposing a big data solution. Their pioneering work established a good foundation for the work of this paper.

To summarize, both qualitative and quantitative evaluations of driving style are required for a more comprehensive analysis of drivers and their driving trips from a large volume of natural driving data. When solving the driving style evaluation problem with unsupervised learning, clustering methods are preferred for identifying driver styles, but the importance of input features was not considered in previous studies. Moreover, quantitative evaluation in an unsupervised manner has seldom been considered for analyzing driver performance by driving trips. In addition, big data statistics offer new technology for discovering and quantifying driving patterns. This motivated our work to combine driver style recognition and trip-level scorecarding for more accurate and dynamic driving performance evaluation.

3. Dataset

In this work, similar to Umberto et al. (2019), we focus on the analysis of driving performance based exclusively on the controller area network (CAN) bus¹ signals and ignore external factors (e.g., weather, traffic conditions). A logistics company located in Jining city, Shandong Province, agreed to participate in the study under privacy constraints. The type of participating vehicle is a heavy truck manufactured by Dongfeng Commercial Vehicle (DFCV), and the onboard Telematics Box² (T-Box) can gather CAN bus signals and send them to a cloud

Table 2

Truck configuration, summary of the dataset and signals used in this study.

Truck configuration	
Truck brand	DFCV Tianlong
Max speed	110 km/h
Max torque	2000 N m
Max torque speed	4000 rpm
Max weight of traction	40 Ton
Idling range	(0, 700) rpm
Driving trip summary	
Number of vehicles/drivers	53
Road type	Mainly highway; the speed limit for trucks is 90 km/h
Valid trips	32,135
Average trips per truck	606
Minimum trip mileage	3 km
Selected signals	
Engine torque and speed	To detect idling or ignition state
Speed, brake pedal position (%)	To detect accelerating/braking events
Bearing (deg)	To detect turning events
Mileage (km)	To calculate trip distance
Instantaneous fuel consumption (mL)	To sum trip fuel consumption

server via a cellular network. The data acquisition frequency is 1 Hz. Detailed information about the vehicle and dataset can be observed in Table 2.

As reported by the fleet manager, vehicles were moving between the surrounding cities once the delivery tasks were assigned. Notably, apart from the destination, the drivers were not given any instruction on the route they had to drive or the speed or behavior they had to follow during the driving. Thus, they were driving freely in a completely uncontrolled and open driving environment. This will result in different driving performances even for the same driver, making trip-level performance evaluation necessary.

3.1. Data description and preprocessing

The data used in this study were collected from January 1, 2019, to March 31, 2019. The dataset is obtained from different paths, but the proportion of highways is relatively large to save time and to avoid entering the central urban area. The dataset is first preprocessed to handle outliers and missing records and then segmented into trips by detecting whether the engine is down for more than 30 min. In addition, data with consecutive values missing for more than 1 min in the time sequence are also regarded as belonging to two different driving trips. Then, trips whose distances are less than 3 km are discarded. A summary of driving trips can be found in Table 2. The selected signals are also listed with their main purposes.

Since records that contain empty values are rare, we simply discard them. Vehicle speeds beyond 110 km/h are ignored, and speeds beyond 90 km/h are considered overspeeds (overSpd). The absolute values of acceleration greater than 20 km/h/s are ignored. Harsh acceleration (harshAcc) and harsh deceleration (harshDec) events were triggered when the speed change rate reached ± 3 km/h/s. The valid bearing angle is within 0 to 360 degrees, and a pairwise change rate greater than 50 deg/s with speed over 40 km/h is treated as a harsh turn (harshTurn). The 99th percentile (2.5 L) of instantaneous fuel consumption readings is used to filter noisy values. Furthermore, mileage signals have no outliers.

3.2. Extracting trip-level features

In fleet management, trip-level rather than driver-level driving performance evaluations are of higher concern since the driver's performance will vary with the situations. An investigation of driver behavior variables (Zfnebi et al., 2017) has shown that speed, acceleration/deceleration and braking are variables most commonly used in recent studies, and they are considered priority variables. Notably,

¹ https://en.wikipedia.org/wiki/CAN_bus.

² <https://en.wikipedia.org/wiki/Telematics>.

Table 3

The extracted metric features representing a driving trip. The features' qualitative effects on driving performance, as reported by the fleet manager, are also listed.

Metric features	Effects
Behavior features	
Frequency of harshAcc per mile	—
Frequency of harshDec per mile	—
Frequency of harshTurn per mile	—
Frequency of overSpd per mile	—
Ratio of idle time	—
Statistical features	
Mean speed (meanSpd)	o
Standard deviation of speed (stddevSpd)	—
Mean acceleration (meanAcc)	—
Standard deviation of acceleration (stddevAcc)	—
Mean deceleration (meanDec)	—
Standard deviation of deceleration (stddevDec)	—
Standard deviation of bearing (stddevBearing)	—

—: Higher value leads to worse performance
o: Either higher or lower value leads to worse performance

**Fig. 1.** Heatmap of behavior features from 20 randomly chosen trips.

detected aggressive driving events are highly representational or informative for measuring driving performance. As a result, we selected the key metrics shown in Table 3.

In Table 3, the behavior feature category addresses drivers' aggressive behaviors, which are quite interpretable for good or bad performance. For example, applying brakes too hard can be a sign that a driver is not paying attention or might be tailgating. Letting a truck idle in a driveway (or on a highway) can greatly reduce the gas mileage. The first 5 driving events are detected in real time with predefined rules. Then, we count the numbers of the first 4 types of driving events, which are divided by the trip distance, analogous to the commonly used indicator, liters per kilometer, which is empirical and practical in industrial applications. The ratio of idling is the total idling time divided by the total running time of the trip. The first 5 metric factors are normalized to ensure that all trips are compared on the same scale. To visualize this vectorized result, Fig. 1 shows the heatmap of normalized aggressive behavioral features from 20 randomly selected trips. The driving performance for the 12th trip is more aggressive than that of the 3rd trip.

Movement statistics measure, at a global level, the skill with respect to how the driver controls the longitudinal and lateral movements of the vehicle. For the 'average speed' factor, since the most economical speed is neither too high nor too low, we treat it as a neutral feature, which means that the greater the deviation from the average speed or the economical speed is, the worse the performance. For other statistical features, small values show smooth changes in vehicular states, which usually correlate with better performance.

These features provide accurate information about the way in which the drivers are driving. One will have accurate data to quantify a driver's performance and prepare a report on the driver's performance.

Table 4

Symbols and notations.

Symbol	Definition
D	Driver set
R	Trip set
$\mathbf{x}$	Any column vector
X	A matrix or a set of vectors
$X_{\cdot j}$	j th column vector of X
$X_{i \cdot}$	i th row vector of X
$P(\cdot)$	Probabilistic distribution
$f(\cdot)$	Probability density function (PDF)
$F(\cdot)$	Cumulative density function (CDF)
μ	Expectation or mean
σ or Σ	Variance terms
π or w	Weight terms

Moreover, measuring features enables achievement of the goal of performance evaluation to reduce aggressive driving behavior and keep trucks under smooth control.

4. Methodology

4.1. Problem definition

The driver group is assumed to contain n drivers, in which the i th driver d_i is supposed to have produced n_i historical trips. We define these trips as $R^i = r_1^i, r_2^i, \dots, r_{n_i}^i$. Then, the problem is as follows: given any driver's newly finished trip r_j^i , how can the driving performance results collected during that trip be evaluated compared to the historically collected R^D ?

Given that the varying-length trip is represented as a fixed-length vector $\mathbf{x}$ containing m features, we later propose a scoring model to help fleet managers transform their qualitative and subjective evaluations of this trip into quantitative scores. However, as mentioned, the challenge is that without already scored samples as training targets, the traditional supervised learning method is not applicable for building a regression model. According to the central limit theorem (Montgomery and Runger, 2010; Rouaud, 2013) and business experience, only a few drivers will likely display very bad or good performances; therefore, we formulated the trip-level quantitative evaluation as a *scorecard model whose output must follow a normal distribution*. By doing so, the fleet manager could easily identify the drivers with the worst or best performances for either driver correction or rewarding.

Additionally, at the driver level, we focus on methods to recognize the driving styles of these D drivers. Driving style is a very important indicator and is a crucial measure of a driver's performance and ability to drive in a safe and protective manner (Yan et al., 2019). In this section, our problem involves recognizing the main driving style of each driver using an unsupervised learning method. The method groups drivers into three levels (calm, normal, and aggressive) according to the level of aggressiveness (low, medium and high, respectively). Most of the existing clustering algorithms are capable of solving this problem, but the challenge is to learn the driver-level representation over the driver's numerous trip feature vectors while considering the feature importance to achieve better clustering results.

4.2. Algorithm framework

Before presenting our framework, some key symbols and notations that are used in this work are defined in Table 4. A lowercase letter in bold denotes a vector, and an uppercase letter denotes a matrix containing a set of vectors. Other common statistical variables are also defined.

To solve the two problems, a block diagram of the proposed driving performance evaluation framework is shown in Fig. 2. At the beginning, related events and sensory signals are obtained in real time and normalized for each driving trip. Then, the feature engineering process

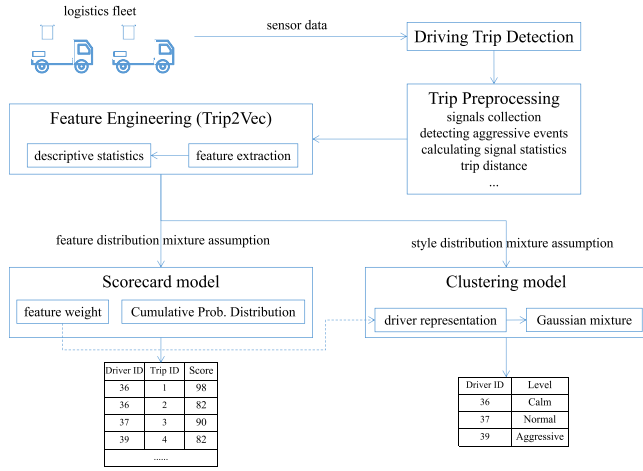


Fig. 2. Block diagram of the proposed driving performance evaluation framework.

commences, in which metric features are constructed to produce fixed-length vectors for all variable-length trips, and descriptive statistics for each feature are obtained. Finally, two branches are drawn: one for a scorecard model, which gives a reasonable score for previous and future trips, and the other for introducing the clustering model, which divides the drivers into discrete groups and gives the corresponding style definitions aided by the estimated scores. The detailed implementation is provided later.

4.3. Trip-level quantitative evaluation

4.3.1. Formulation of the scorecard

Fleet and safety managers can use a scorecard to identify the best drivers in the fleet so that those drivers can be rewarded. Among the advantages of using scorecards for driver behavior management, we can highlight their flexibility, since they can be designed for the needs of a specific fleet, and the objectivity of their measurement, which provides a fair rating for all drivers.

Without the help of target labels, the scorecard is built by comparing a trip to the trip population to determine its score. This idea requires creating an objective scoring mechanism to provide each trip with an evaluated score, effectively measuring how skilled (risk-prone) a driver is relative to the general driver population. This score ranges from 0 to 100, and a higher score indicates a better driving performance of r_j^i within R^D . The score is obtained with Eq. (1):

$$\begin{aligned} \text{Score}(r_j^i) &= S(\psi(r_j^i) | \psi(R^D)) \\ &= S(\psi(r_j^i) | \psi(R^i) \cup \psi(R^{D/d_i})) \end{aligned} \quad (1)$$

where S is a scorer/regressor, R denotes the set of total trips produced by D drivers with the same type of vehicle and ψ is the Trip2Vec transformer, which maps the originally gathered driving data with varying length to the trip-level feature space required by S , namely, $x = \psi(r)$.

Before proposing the scorecard model, we observe the histogram of each feature. Fig. 11 provided in the Appendix displays the distributions of all the features. As shown, all the occurrence numbers of aggressive driving behaviors are exponentially distributed. This means that the majority of trips have low risk levels. In contrast, all the histograms of statistical features display bell-like shapes, indicating a (skewed) normal distribution. According to this observation, we are able to model the distribution of each feature with either exponential or normal PDFs. Moreover, we concluded that the trip-level representation is a mixture of feature-level distributions with the following definition.

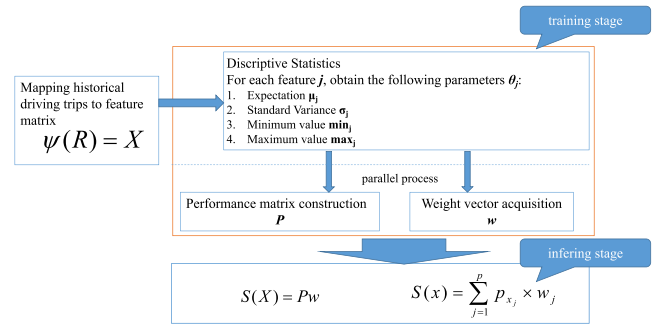


Fig. 3. Pipeline of the trip scoring system.

Definition 4.1 (Feature Distribution Mixture). Each feature has a common parameterized PDF denoted by $P(\cdot) = f(\cdot; \theta)$, and the **big trip data** enable the estimation of PDF parameters for each feature. More importantly, the features are considered independent components, and all features together shape the distribution of the trip representation.

This definition can be mathematically defined with Eq. (2). Generally, considering that the dimension of the feature space is m and n trips are collected, then the mixture distribution of $X_{i=1, \dots, n; j=1, \dots, m}$ is the weighted integration of the individual feature distribution, where θ_j denotes the parameters that define the form of the specific PDF of the j th feature, and w is a vector of m weights that sums to 1.

$$\begin{aligned} P(X_{*j=1 \dots m}) &\approx \int_{j=1}^m w_j \cdot P(X_{*j}) \\ &= \int_{j=1}^m w_j \cdot f(X_{*j}; \theta_j) \end{aligned} \quad (2)$$

4.3.2. Scoring with feature distribution mixtures

As illustrated in Fig. 3, the pipeline of this scorecard begins with each trip being transformed into a feature set (vectors), and then the process is guided by the arrows to the training stage. The main function of this stage is to estimate each feature's statistics (the minimum, maximum, average and standard variance) to approximate the corresponding PDF form for each metric feature. Then, the feature vectors are used in parallel to produce a performance matrix with the PDFs as well as the feature weights obtained from information theory. In the inference stage, the scoring of the training data is calculated by matrix multiplication on the left, and the scoring of the newly input trip data is calculated by the equation on the right.

Note that the cumulative density of any feature distribution is between 0% and 100%, and a traditional scoring system is always supposed to output a score between 0 and 100, with a higher score indicating better driving performance. We then establish the correlation between the scoring criteria and the feature CDF based on the following assumption.

Definition 4.2 (Scoring Criteria). The more aggressive maneuvers a driver performs, the more likely he/she is to be an unsafe driver. A driver who has zero instances of harsh turns, harsh accelerations, speeding or harsh braking over some distance is more likely to be classified as a safe driver. In contrast, we can assume that a driver who makes multiple harsh movements or exhibits higher moving vibrations over the same distance should be penalized with a low score.

The score of each feature based on Definition 4.2 should be closely linked with the cumulative probability density of the feature. The following can be observed from Fig. 11: (1) for behavior features that have exponential distributions, a lower cumulative probability density contributes to a higher performance score, (2) for the special case of feature 'meanSpd' that has the 'o' effect, the optimal driving performance lies at the peak when its value is the expectation μ , and (3) for the other statistical features that have normal distributions, we

assume that a smaller cumulative density implies that the driver is more skilled.

This definition contributed to the method for training a scorer S given only X . From the perspective of probability, this scorecard is modeled as a mixed function resulting from the weighted mixture (sum for discrete variables) of the feature distributions of the following form:

$$P(S|X_{*j=1\dots m}) = \sum_{j=1}^m w_j P(X_{*j}) \quad (3)$$

Specifically, we define the scoring function as the following weighted sum of the CDF-related term:

$$S(x) = \sum_{j=1}^m w_j \cdot \begin{cases} (1 - F_j(x_j)) & j \text{ with '−'} \\ (1 - F_j(|x_j - \mu_j|)) & j \text{ with 'o'} \end{cases} \quad (4)$$

where the subscript j represents the j th feature in the feature vector x . w_j and F_j are the associated weight and CDF, respectively. For a feature that has a '−' effect, we only estimate its CDF value, while for a feature that has a 'o' effect, we calculate the cumulative density of $(|x_j - \mu_j|)$. Then, we subtract all CDF values from 1 and perform a weighted sum across all features.

As the above equations demonstrate, individual scores are associated with all the features to show the aspects in which a driver performs well or poorly, and then these individual scores are input into our linear unit to determine an overall driving safety score. These individually scored vectors form the mentioned performance matrix P , which is obtained with Eq. (5), where the element P_{ij} indicates a score under the criterion proposed by the j th feature.

$$P_{ij} = \begin{cases} (1 - F_j(X_{ij})) & j \text{ with '−'} \\ (1 - F_j(|X_{ij} - \mu_j|)) & j \text{ with 'o'} \end{cases} \quad (5)$$

With the scores given by the metric features and the associated safety levels, this method can differentiate the drivers labeled as aggressive by the risk intensity. This is a naive approach because it ignores the dependency among the metric features; however, it helps to make the model easily explainable, similar to ranking the students of a class by a weighted sum of all course scores.

4.3.3. Feature weight strategy

This section will consider the feature importance. Generally, weights are obtained with two types of methods, i.e., subjective and objective. We suggest the latter method. The subjective method is inspired by information entropy theory, which is a measure of uncertainty. In particular, the entropy weight (EW) method is adopted. It is assumed that features with higher-level information are more important and, therefore, that they have higher weights. Given the input matrix X , the EW method is implemented through the following steps:

- Step 1: Normalize X with Eq. (6)

$$X_{ij} = - \frac{\text{Max}(X_{*j}) - X_{ij}}{\text{Max}(X_{*j}) - \text{Min}(X_{*j})} \quad (6)$$

- Step 2: Calculate the ratio A_{ij} of each X_{ij} under the j th feature by Eq. (7):

$$A_{ij} = \frac{X_{ij}}{\sum_{i=1}^n X_{ij}} \quad (7)$$

- Step 3: The column entropy vector e is obtained through Eq. (8):

$$e_{j=1\dots m} = - \frac{1}{\ln(n)} \sum_{i=1}^n A_{ij} \ln(A_{ij}) \quad (8)$$

- Step 4: The column diverse degree $g_j = 1 - e_j$ is computed.

- Step 5: The individual objective weight of each feature is defined according to Eq. (9):

$$w_j = \frac{g_j}{\sum_{i=1}^m g_j} \quad (9)$$

The larger the value of e_j is, the greater the differentiation degree of feature j is, and the more information that can be derived. Hence, a higher weight should be given to the feature. Although each feature is reasonably accepted by end users in measuring their driving performance and is treated as an independent variable, each feature has unique importance in comparing driving performance trip by trip. That is why we use this objective method to obtain a feature weight vector and then to adjust the importance of the input feature vector.

4.4. Driver-level qualitative evaluation

As mentioned above, the general tendency of driving style recognition at the driver level is to group drivers into classes with interpretable styles. We discuss two ways to obtain a finite number of driving styles, which are safety-related and in reference to the level of aggressiveness (e.g., high, medium and low levels of risk intensity).

4.4.1. Clustering with style distribution mixtures

To cluster drivers, we introduce the probabilistic model based on the following definition:

Definition 4.3 (Latent Style Mixture). The driver population contains several subpopulations separated by the similarity of their vectorized representations, and each group corresponds to one type of driving style. These styles are considered independent and have nonidentical latent distributions. Due to the fluctuations in driving behavior, every driver has a certain probability of belonging to one of these styles, but overall, each driver has a major style tendency.

Unlike Definition 4.1, which focuses on the relationship between features (vertical direction view), this definition addresses the horizontal-direction identification of the subpopulation to which an individual observation belongs. If there are K parametric distributions, then the marginal distribution of a driver representation $z \in \mathbb{R}^m$ is defined with Eq. (10):

$$f_k(z; \Theta) = \sum_{k=1}^K \pi_k f(z; \theta_k) \quad (10)$$

where the weight $\pi_k \in [0, 1]$ satisfies $\sum_{k=1}^K \pi_k = 1$ and Θ are the distribution parameters with $\Theta = \{\pi_1, \dots, \pi_K, \theta_1, \dots, \theta_K\}$. The capability of this model obviously depends on each PDF $f(z; \theta_k)$, which can be associated with any type of distribution.

The latent style mixture assumption (e.g., Gaussian mixture (Miyajima et al., 2007; Wang and Lukic, 2011)) has been proposed in previous literature. Similarly, we model the classification of driving styles as a mixture of K Gaussians. Moreover, we present a weighting strategy for improvement, which extends Eq. (10) to the following weighted Gaussian mixtures:

$$p(x) = f(z; \mu, \Sigma) = \sum_{k=1}^K \pi_k N(w \odot z | \mu_k, \Sigma_k) \quad (11)$$

where $\odot$ refers to elementwise multiplication between two vectors. As shown, the slight difference in the model from a traditional GMM is that we adjust the input vector z with a weight vector.

To determine their optimal values, we use the maximum likelihood estimation method. This method is based on the likelihood function, $p(Z)$, defined as the joint probability of all observations Z , as shown in Eq. (12):

$$p(X) = \prod_{i=1}^N p(X_i) = \prod_{i=1}^N \sum_{k=1}^K \pi_k N(w \odot Z_i | \mu_k, \Sigma_k) \quad (12)$$

The logarithm is commonly applied to each side of Eq. (12), and the negative log-likelihood function can be written by Eq. (13). The parameters Θ can be estimated by minimizing $L(\Theta)$. To accomplish this, we use an iterative method called the expectation–maximization algorithm.

$$L(\Theta) = - \sum_{i=1}^N \log \left(\sum_{k=1}^K \pi_k N(w \odot Z_i | \mu_k, \Sigma_k) \right) \quad (13)$$

Additionally, in a Bayesian setting, the mixture weights and parameters are random variables, and prior distributions are placed over the variables. It is also important to note that GMM implies the K-means algorithm when applied to clustering problems. The latter is a special case of a GMM, specifically, the limiting case when fixing all covariances to be diagonal, equal and having infinitesimally small variances (Press et al., 2007). Instead of small variances, a hard cluster assignment can also be used to show another equivalence of K-means clustering to a special case of “hard” Gaussian mixture modeling. Gaussian mixture modeling can be interpreted as a generalization of the K-means clustering; in contrast, the use of K-means clustering to find the starting points for Gaussian mixture modeling in situations with difficult data has been suggested.

Next, we handle the challenge of how to extract the driver-level features that should be fed into the clustering model. We first show that purely averaging the trip-level feature vectors is not the optimal choice; then, we propose an aggregation with a mixture of mean and median values.

4.4.2. Driver-level representation

As is well known, a driver's driving style can vary depending on the situation, and consequently, variable performance scores can be produced from one trip to the next. It is necessary to extract a driver-level representation from the trip vectors to capture the main style of each driver. A trip can be treated as a view of a certain driver's driving style. However, some trips could be partial reflections or even noise compared to this driver's main style. Extracting driving style information from trip data only at the trip level can be impacted by local factors, such as road shapes and traffic conditions. As shown in Fig. 4, which is a t-distributed stochastic neighbor embedding (t-SNE) plot of trips conducted by drivers 38 and 52, there is a trip from driver 52 located to the far right. This trip could be treated as ‘noise’. Intuitively, the two drivers should have different driving styles, but some of their trips still overlap.

As a result, simply taking the mean vector from all trip vectors as the driver-level representation could be significantly impacted by noisy trips. Taking the median values of each feature could be more robust, especially for behavior features that are exponentially distributed; however, the experimental results presented later in the text show that this strategy does not improve the clustering performance since the distributions of the statistical features are not standard normal distributions (namely, mean $\neq$ median). Instead, we present a mixed method for determining driver-level representation. Suppose the vector for driver d_i is $\mathbf{z}^{d_i} = (z_1^{d_i}, z_2^{d_i}, \dots, z_j^{d_i}, \dots)$, where the j th representation value depends on the distribution of the j th feature, which is obtained with Eq. (14):

$$z_j^{d_i} = \begin{cases} \text{median}(X_{*j}^{d_i}), & X_{*j} \sim E(\lambda) \\ \text{mean}(X_{*j}^{d_i}), & \text{otherwise} \end{cases} \quad (14)$$

5. Distributed implementation

Current fleet management systems face big data challenges in that (1) the large-scale data collected from a fleet of vehicles require massive storage space and better organization for efficient queries and (2) once a driving trip is finished, the scorecard model is expected to predict a score in real time. This requires online preprocessing of the streaming data flow. To apply our proposed framework to real-world cases, in this section, we also propose a big data solution to surmount the above challenges.

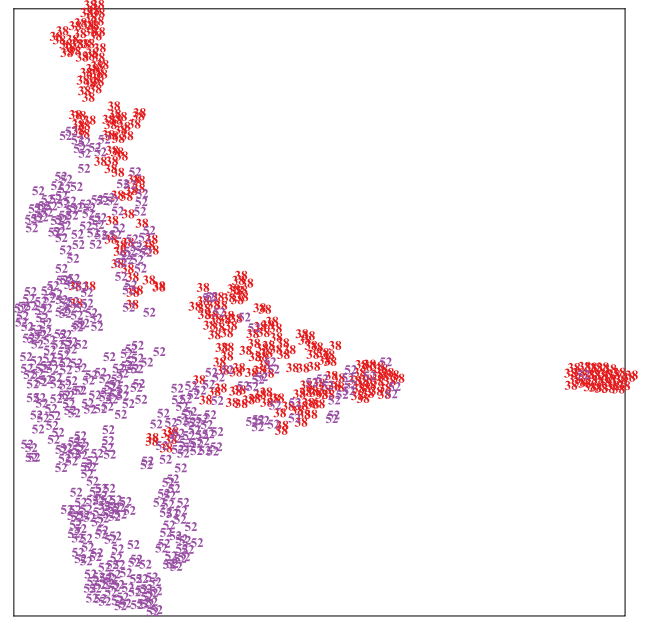


Fig. 4. The t-SNE plot of two groups of trip features from drivers 38 and 52.

5.1. Big data platforms

Big data techniques enable data mining (access and analysis) of vast amounts of in-vehicle parameters to derive models, patterns and results. Therefore, we build a platform integrating our proposed methods to store historical data and to handle real-time streaming sensor data. We choose a simplified technology stack, which includes Apache Kafka,³ Hadoop,⁴ Spark,⁵ Oozie,⁶ and Hive.⁷ Kafka is a message queue tool responsible for receiving the sensor data remotely transmitted by each vehicle. The platform mainly uses Spark structured query language (SQL) and machine learning (ML) to perform data access and model training. Spark streaming is used to handle online signal time-series preprocessing and model prediction. The historical data are stored on a Hadoop distributed file system (HDFS) with built-in Hive tables for easy queries. Then, the Oozie Workflow Scheduler is chosen to schedule specific offline jobs.

5.2. Implementation flow

To implement this solution, we set up a cluster containing three nodes, where each node has 32 GB of memory and eight cores. The historical dataset used in this study, with a total size of 50 GB, was migrated into a Hadoop distributed file system with all the drivers remaining anonymous. Corresponding to Fig. 2, the key implementation flow of our framework is shown in Fig. 5. The whole process consists of three jobs: preprocess and trip detection, feature extraction, scoring model building (including feature weight calculation) and driver clustering. First, one Spark streaming job consumes a batch of signal sequences from the Kafka queue every 5 s, and it initially executes two tasks. The first task filters outliers and detects driving events with predefined rules and updates the details of each type of behavior event into the *driving behavior* table. The second task detects whether the current journey has ended. Once finished, the system assigns another unique ID for the next new trip. For the finished trip, the vehicle

³ <http://kafka.apache.org/>.

⁴ <http://hadoop.apache.org/>.

⁵ <http://spark.apache.org/>.

⁶ <https://oozie.apache.org/>.

⁷ <https://hive.apache.org/>.

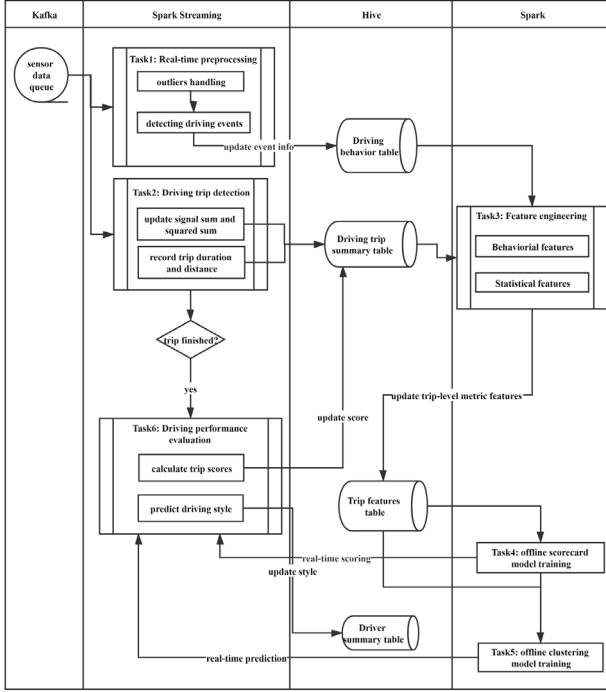


Fig. 5. The real-time driving performance evaluation system implementation flow.



Fig. 6. A typical app for evaluating driving performance that includes a style label, an overall score and detailed scores.

identity number, start/end time, duration, and other information about the trip are inserted into the *driving trip summary* table, where the sum and squared sum of related signal values are updated in real time to calculate the mean and standard variance.

Assuming that the aforementioned two Hive tables have accumulated enough data, the offline Spark job loads the data from both tables. During the feature engineering stage, the numbers of aggressive driving events are normalized, and the signal statistical features are computed to represent each trip, as mentioned in Section 3.2. These trip-level features are stored in the *driving trip feature* table, which will be fed into the offline scorecard model training job. Task 4 first obtains the statistics of each feature, including the maximum, minimum, mean, and standard deviation. Additionally, the EW and performance score

of each feature are also calculated in parallel, as displayed in Fig. 3. These feature statistics and weights together fit a scoring model.

The following is the driver clustering model training job, which first aggregates each driver's trip vectors into a single vector at the driver level. Finally, the representation vector set of all the drivers, as well as the feature weights, are used to train an optimal clustering model for driver style recognition. The best clustering model is chosen through a comparative experiment that is presented later. If the above models have been trained and stored in HDFS, task 6 of Spark streaming is to obtain the relevant information of the current trip, use the two models to score and identify the style, and update the score in the *driving trip summary* table and the style in the *driver summary* table for later presentation.

The main contribution of this research is the distributed scoring algorithm. Algorithm 1 shows the distributed implementation of our proposed scorecard model programmed with Spark. In addition, we propose two weighted clustering methods by adding the EWs of features to K-means and GMM provided by Spark mllib. All the algorithms render our framework applicable to large-scale fleet management and enable satisfying real-time prediction requirements.

Algorithm 1 Pseudocode for distributed training and prediction for the scorecard model

Input: Trip feature DataFrame/RDD X

Output: Scores $S(X)$

Model Fit Procedure:

Obtain the feature statistics (min, max, mean, std) map, denoted as $\{feaMinMap, feaMaxMap, feaMeanMap, feaStdMap\}_{j=1,...,m}$

for each feature j do

Normalize X_{*j} using Eq. (6) assisted by $feaMinMap_j, feaMaxMap_j$

Compute $s = X_{*j}.sum()$

Compute $scalar = -1.0/\log(X.count())$

Set $e_j = 0$

for each $x \in X_{*j}$ in parallel do

Compute $a = x/s$ with Eq. (7)

Compute $e = a * \log(a + 1e^{-6})$ with Eq. (8)

Set $e_j = e_j + e$

end for

Compute $g_j = 1 - e_j \times scalar$

end for

Calculate weight w with g by Eq. (9)

Model Prediction Procedure:

for row vector X_{i*} in parallel do

Set $S(X_{i*}) = 0$

for $X_{ij} \in X_{i*}$ in parallel do

Compute P_{ij} using Eq. (5) and $feaMeanMap, feaStdMap$

Set $S(X_{i*}) = S(X_{i*}) + P_{ij} \times w_j$

end for

end for

Return $S(X)$

5.3. Interpretable presentation

Finally, this system offers a mobile app as a user interface to display the evaluation outputs, as shown in Fig. 6. Generally, machine learning algorithms provide the style types and trip scores in a black-box manner. However, these opaque results are insufficient to convince managers and drivers of poor or good performance. As a result, detailed evaluations are also required for interpretability. In Fig. 6, however, detailed scores could be provided to help managers and drivers gain insights into the composition of this total score.

Taking Fig. 6 for instance, suppose driver 36 performed 7 speeding violations, 7 harsh accelerations, 5 harsh brakes, and 5 harsh turns per kilometer during his/her driving trip. The next step would be to

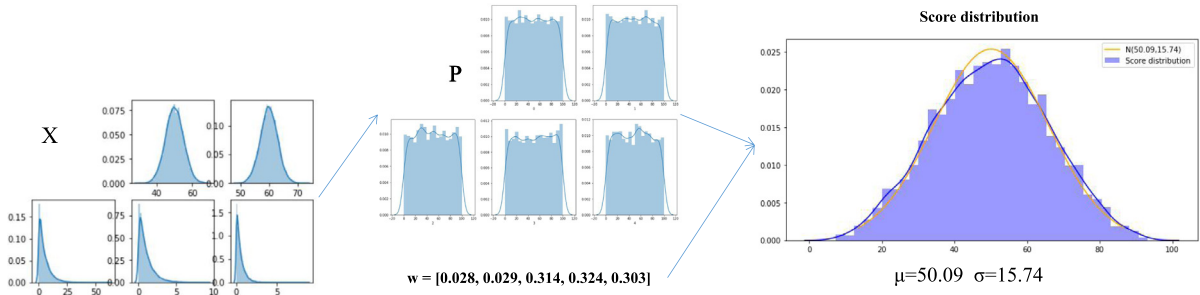


Fig. 7. Illustration of scorecard modeling on synthetic data.

calculate this driver's CDF values for each feature. Let us assume that we obtained a CDF vector of [0.88, 0.81, 0.32, 0.32] for speeding, accelerations, brakes, and turns, respectively. Assuming that the decided weights for each of the behaviors are 0.3 for speeding and harsh turns and 0.2 for brakes and accelerations, the final score would be $1 - (0.4 * (0.88 + 0.81) + 0.2 * (0.32 + 0.32)) = 19.6\%$. We could further explain this result by highlighting that the main reason this driver obtained this score was because he/she was at high percentiles: 88% and 81% within the groups of speeding and harsh accelerators, respectively. Moreover, suggestions based on the features' scores could also be provided to help drivers improve their driving performance.

6. Experiments

Our empirical evaluation of the proposed framework is as follows: first, synthetic data are evaluated without the influence of noise, and then real-world data are evaluated to enable a comparison and robustness check. With these data, we attempt to demonstrate through experiments that the two models are reasonable and practical for implementation by end users.

6.1. Model validation settings

The first step in building the scorecard model is to determine the PDF of each feature. The PDF could take any permitted form, but as previously discussed, Fig. 11 suggests either an exponential or normal distribution with some skewness. Hence, we arbitrarily define the behavioral features to have the exponential form as follows: $x \sim E(\lambda)(\lambda = 1/\mu)$. Other statistical features follow the normal form, as follows: $x \sim N(\mu, \sigma^2)$. The two distributions' PDFs are shown in Eqs. (15) and (16). The key point of the scoring model is to determine whether the output score distribution conforms to the normal distribution trend, among which the low score and high score samples are effectively distinguished. The key consideration of driver clustering effect verification is whether the clustering algorithm that considers the importance of features in a mixed representation is better. We choose GMM, K-means, Bayesian Gaussian mixture (BGM) and Spectral clustering as the candidate methods.

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0, & x < 0 \end{cases} \quad (15)$$

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2} \quad (16)$$

6.2. Scoring model validation

According to the central limit theorem, most scoring systems approximately follow certain normal distributions. Since the target scores over all trips are not available, this section mainly adopts qualitative analysis on noise-free synthetic data and then on the real trip vectors mentioned in Section 3.2.

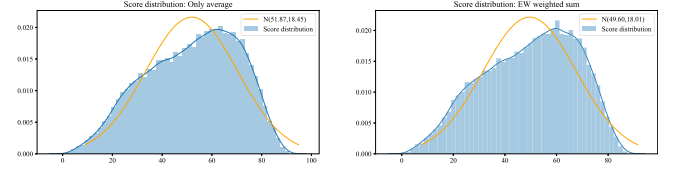


Fig. 8. The histogram plots of the trip-level overall scores. For comparison, the left histogram shows the average scores, treating each feature as equally important.

6.2.1. Test on toy data

We arbitrarily generate a matrix X (considered trip vectors) with 5000 rows and 5 columns; the first two columns are sampled from two normal distributions, and the other three are sampled from exponential distributions. We only use the recommended EW method to train our scorecard model to quickly prove that variables with other PDF forms (such as normal distributions) can also be integrated, and the output of the model has the same normal shape as many other scoring systems.

Fig. 7 illustrates the data distribution and the scoring process. After the performance matrix P is constructed, the middle part displays the distribution of each column in P , as well as the calculated EWs. The values in the weight vector show that greater importance is assigned to the three exponentially distributed variables. In Fig. 7, the histogram plot of the final score vector is displayed as a sample set represented by the normal distribution $N \sim (50.09, 15.74)$ (the H_0 hypothesis). Next, Kolmogorov–Smirnov (K–S) and Anderson–Darling (A–D) tests are conducted to prove that the H_0 hypothesis cannot be clearly rejected. Such goodness-of-fit statistical tests show that our proposed scorecard model is able to produce normally distributed scores over a homogeneous population.

6.2.2. Test on real data

For the real driving data provided by the logistic fleet, the offline feature engineering job is first executed to produce a trip vector set. Then, the corresponding scorecard model is trained under different settings. The EWs of all the metric features, in accordance with the order in Table 3, are 0.075, 0.071, 0.071, 0.076, 0.070, 0.119, 0.120, 0.079, 0.073, 0.089, 0.075, and 0.082. As can be observed, the two features 'meanSpd' and 'steddevSpd' are of higher importance, while the feature 'ratio of idling' per trip has the lowest weight.

The histogram of the trip-level entropy weighted scores is plotted in Fig. 8, where the kernel density estimated curve and corresponding standard normal curve are provided. For comparison, the histogram of simply averaged scores is also presented. Both distributions still maintain the normal distribution shape, where the scores of most trips are concentrated in the middle area and those with lower or higher scores are in the minority. It is difficult to determine which is better, but we show that the EW method helps improve the clustering performance in a later experiment.

Next, we select three pairs of trip vectors with the predicted scores located at the left-most, middle and right-most positions in Fig. 8 for

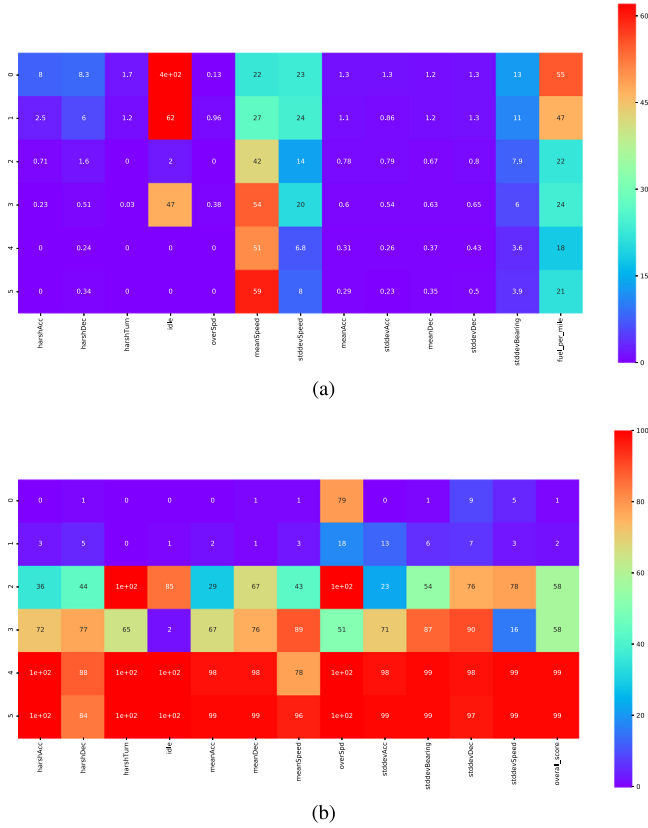


Fig. 9. Heatmaps of the three sampled trip pairs with low, medium and high scores. The upper figure displays the feature vectors plus fuel consumption per mile, while the bottom figure shows the corresponding performance and overall scores. In both figures, the first two rows are the two trips with the lowest overall scores but which yield the highest fuel consumption. The last two rows are the two trips with the highest scores and lower fuel consumption.

qualitative analysis. Fig. 9(a) first displays the heatmap of six sampled trip feature vectors plus their calculated fuel consumption per mile. Fuel consumption is a critical factor in measuring driving performance since features such as aggressive acceleration, idling or non-eco-speed consume more fuel energy. Fig. 9(b) is the heatmap of the feature scores and final scores. By comparing the two subfigures, it can be proven that the trips with ‘meanSpd’ approximating an economical speed and other features having low values are given high scores. Moreover, we find that trips with low scores are more likely to waste more fuel energy, which reasonably demonstrates the effect of our scorecard model.

Then, business feedback is conducted by requiring a fleet manager to review a report. This report includes the scoring results of all trips, the sampled heatmap in Fig. 9, and Table 11 in the Appendix showing the relationship between drivers’ style and their scores and fuel consumption. The fleet manager, based on his familiarity with the members, subjectively evaluated our scoring method and gave positive feedback because it can effectively rank all driving trips, help them identify poor or better driving processes and help them make intelligent decisions.

6.3. Driver clustering validation

For the driver style classification problem, we define $K = 3$, which indicates dividing drivers into three levels by their economical and safe driving performance. The semantic labels are calm, normal and aggressive and are widely accepted today. Generally, the number of drivers labeled calm or aggressive should constitute a minority compared to those with normal styles. All the clustering algorithms (GMM,

Table 5

Candidate driver-level vectors (inputs) for clustering models.

Without EW	With EW
Mean	EW-Mean
Median	EW-Median
NE	EW-NE
Mixed	EW-Mixed

Table 6

Comparison of the clustering performance evaluations on the toy data.

Input	Silhouette	Calinski_Harabasz	Davies_Bouldin
K-means			
Z	0.299	2461.639	1.100
$w \odot Z$	0.350	5390.635	0.936
GMM			
Z	0.293	2387.765	1.116
$w \odot Z$	0.342	5249.9095	0.944
BGM			
Z	0.293	2380.529	1.117
$w \odot Z$	0.342	5244.028	0.944
Spectral			
Z	0.180	1049.322	1.868
$w \odot Z$	0.172	2178.842	1.723

K-means, BGM, Spectral) are properly tuned to output three groups, whose corresponding semantic labels will later be determined by the average score of the trips within each group.

6.3.1. Candidate driver representations

For driver style estimation, the driver-level vectors, denoted by Z , should be first extracted according to Section 4.4.2. In this section, first, four candidate aggregating methods are proposed. The first two typical approaches are to take the mean and median vectors from each driver’s trip-level feature vectors. The third method is the mixed representation method mentioned in Section 4.4.2. The last method is to take the normalized sum of all the trip-level feature vectors in a similar way to the approach in Dong et al. (2017). Given R^i and the extracted trip features $\{x_k, k \in \{1, \dots, n_i\}\}$, the normalized extraction (NE) of the driver-level driving style feature representation is defined as follows:

$$z_{d_i}^j = \frac{\delta_{d_i}^j}{\max_j \delta_{d_i}^j} \quad (17)$$

where $\delta_{d_i}^j = \sum_{k=1}^{n_i} x_k$ is the vector sum, and $\delta_{d_i}^j$ denotes its j th feature. This representation is used for comparison with our proposed methods.

Additionally, considering whether these representation methods are adjusted with the EW, we finally obtain eight types of clustering inputs, which are listed in Table 5. Notably, to ensure a fair comparison of clustering methods that were fit on $w \odot Z$ against those that were fit on Z , the labels predicted by weighted clustering models will be validated on input Z as well. By doing so, the validated results are able to show whether better clusters are defined by our methods.

6.3.2. Test on toy data

This section serves to validate the effect of the proposed feature weight strategy. We still consider the synthetic data used in Section 6.2.1 but treat each row as a driver-level feature vector. To evaluate the clustering performance of the data without labels, three methods are adopted, namely, the **silhouette coefficient**, the **Calinski-Harabasz** index and the **Davies-Bouldin** provided in scikit-learn.⁸ A higher score with the first two and a lower score with the last one indicate a model with better defined clusters.

⁸ <https://scikit-learn.org/stable/modules/clustering.html#clustering-performance-evaluation>.

Table 7

Descriptive statistics of the scores within each cluster using weighted input from the toy data.

Methods	Statistic	Group 1	Group 2	Group 3
Weighted K-means	Count	1449	2215	1336
	max	51.28	61.51	93.19
Weighted GMM	Count	1726	2031	1243
	max	57.82	61.62	93.19
Weighted BGM	Count	1745	2020	1235
	max	57.82	61.62	93.19
Weighted spectral	Count	1646	1737	1637
	max	54.40	91.59	93.20

Table 8

Performance evaluation comparison of driving style clustering on real data without weighting.

Input	AR	AMI	Hom	Com	FM
K-means					
Mean	0.1641	0.1844	0.2209	0.4923	0.6007
Median	0.1641	0.1844	0.2209	0.4923	0.6007
NE	0.0435	0.0541	0.0899	0.0899	0.3670
Mixed	0.1641	0.1844	0.2209	0.4923	0.6007
GMM					
Mean	0.2442	0.2749	0.3065	0.5824	0.6281
Median	0.0762	0.0767	0.1153	0.3982	0.5815
NE	0.0921	0.0819	0.1166	0.1182	0.4047
Mixed	0.2506	0.2755	0.3065	0.6017	0.6328
BGM					
Mean	0.2442	0.2749	0.3065	0.5824	0.6281
Median	0.0747	0.0740	0.1153	0.3859	0.5804
NE	0.0765	0.0738	0.1088	0.1094	0.3915
Mixed	0.2700	0.2889	0.3192	0.5286	0.6249
Spectral					
Mean	0.1013	0.1634	0.1481	0.4220	0.5851
Median	0.1321	0.2122	0.1832	0.4614	0.5926
NE	0.0229	0.0385	0.0745	0.0815	0.3796
Mixed	0.1321	0.2122	0.1832	0.4614	0.5926

Table 6 shows a comparison of the four clustering algorithms using both types of inputs. For input Z , K-means produces the best results (the 1st row), which are slightly better than GMM and BGM. Spectral is far less effective than the other three algorithms. It also performs worst for input $u \odot Z$. Another significant finding is that K-means, GMM and BGM using weighted input show considerable improvement with respect to grouping Z .

Next, we obtain the scores for each sample in Z according to Section 6.2.1 and then group them based on the clustered labels predicted by these weighted models. Table 7 lists the number of samples and maximum score within each group. Because the scores follow the normal distribution, the observation of two such statistics can further verify whether the clustering algorithms can effectively distinguish the samples with high, middle and low scores. By observing the maximum group-level scores, weighted GMM and BGM seem to encounter difficulty in distinguishing between group 1 and group 2, while Spectral fails to distinguish between group 2 and group 3. Weighted K-means stands out as a more reasonable grouping strategy since the maximum scores of the three groups are farther apart.

6.3.3. Test on real data

For the 53 selected drivers, we first invite the fleet manager to provide the drivers' main styles according to their historical driving performance. All the drivers' labels are provided in Table 11, along with their number of trips, score statistics and fuel consumption statistics. Given the style labels and eight types of driver representations, we evaluate the four clustering candidates with the following five evaluation indexes: Adjusted Rand (AR) score, Adjusted Mutual Info (AMI) score,

Table 9

Performance evaluation comparison of driver clustering on real data with weighted input.

Input	AR	AMI	Hom	Com	FM
K-means					
EW-Mean	0.1641	0.1844	0.2209	0.4923	0.6007
EW-Median	0.1641	0.1844	0.2209	0.4923	0.6007
EW-NE	0.0435	0.0541	0.0899	0.0899	0.3670
EW-Mixed	0.1641	0.1844	0.2209	0.4923	0.6007
GMM					
EW-Mean	0.1306	0.1446	0.1832	0.4546	0.5915
EW-Median	0.2739	0.2862	0.3167	0.5364	0.6282
EW-NE	0.0919	0.0815	0.1164	0.1184	0.4056
EW-Mixed	0.2739	0.2862	0.3167	0.5364	0.6282
BGM					
EW-Mean	0.2442	0.2749	0.3065	0.5824	0.6281
EW-Median	0.2739	0.2862	0.3167	0.5364	0.6282
EW-NE	0.0720	0.0687	0.1039	0.1050	0.3899
EW-Mixed	0.2739	0.2862	0.3167	0.5364	0.6282
Spectral					
EW-Mean	0.1013	0.1638	0.1481	0.4220	0.5851
EW-Median	0.1321	0.2122	0.1832	0.4614	0.5926
EW-NE	0.0229	0.0385	0.0745	0.0815	0.3796
EW-Mixed	0.1321	0.2122	0.1832	0.4614	0.5926

Table 10

Comparison of different types of inputs producing the highest scores (%) for all clustering methods.

Input type	AR	AMI	Hom	Com	FM
Mean	24.42	27.49	30.65	58.24	62.81
Median	16.41	18.44	22.09	49.23	60.07
NE	9.21	8.19	11.66	11.82	40.47
Mixed	27.00	28.89	31.92	60.17	63.28
EW-Mean	24.42	27.49	30.65	58.24	62.81
EW-Median	27.39	28.62	31.67	53.64	62.82
EW-NE	9.19	8.15	11.64	11.84	40.56
EW-Mixed	27.39	28.62	31.67	53.64	62.82

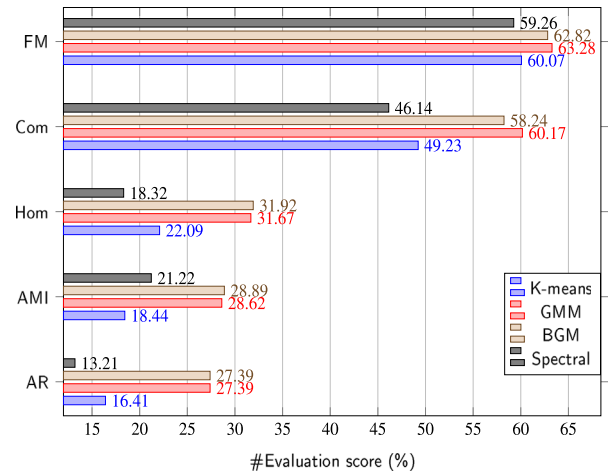


Fig. 10. Comparison of clustering models producing highest scores (%) based on all types of inputs.

Homogeneity (Hom) score, Completeness (Com) score, and Fowlkes Mallows (FM). For all the evaluation methods, a higher score yields better clustering performance.

For the four clustering inputs without multiplying the EW vector, Table 8 lists the scores of the five evaluation methods by comparing the ground truth to the clusters predicted by K-means, GMM, BGM and Spectral. The performance evaluation results for the other four weighted inputs are listed in Table 9. The highest score of each index

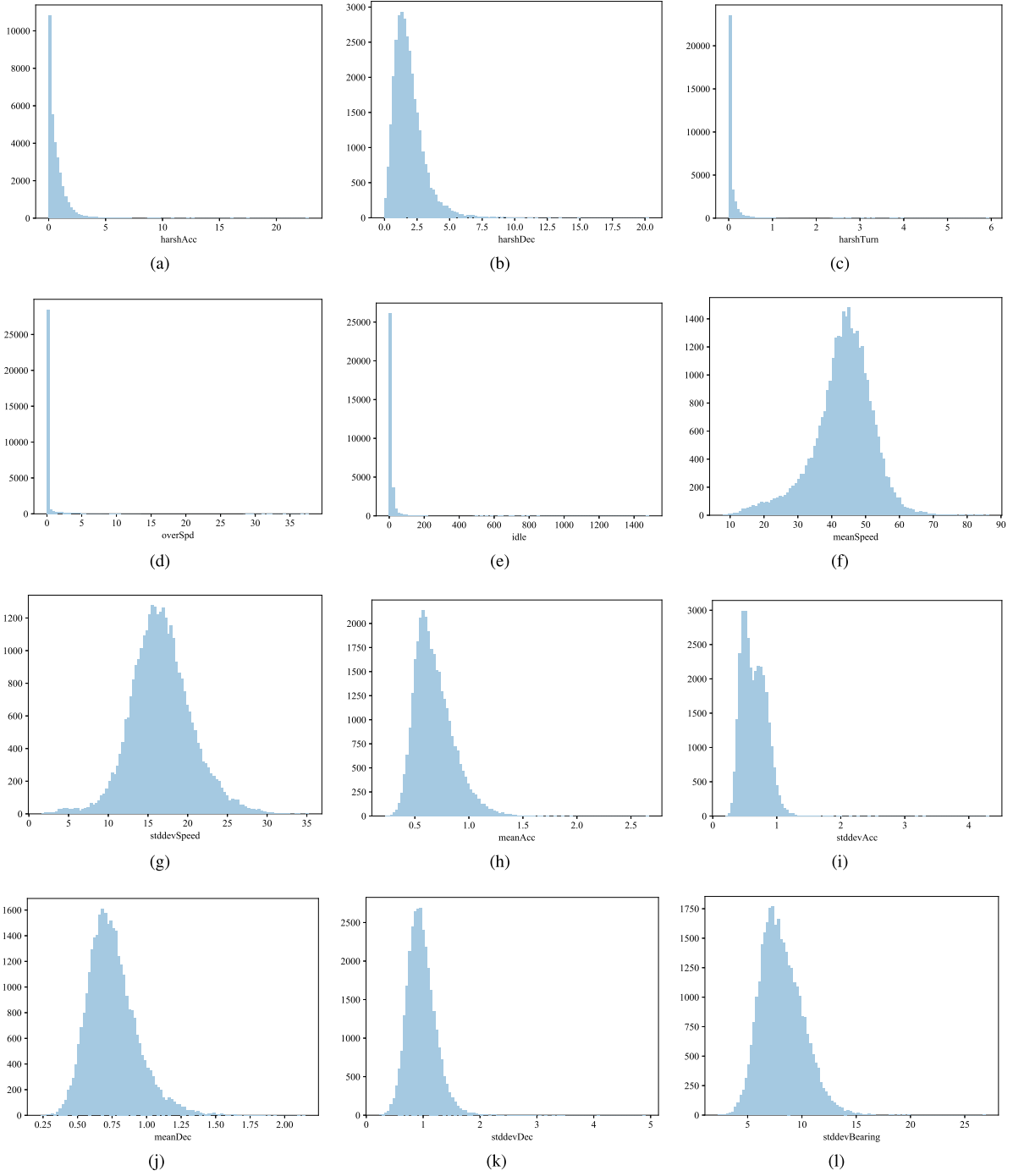


Fig. 11. The histograms of all the trip-level features, where the first five features show exponential-like distributions and the rest are Gaussians. This observation supports Definition 4.1 and the choice of PDF functions in Section 6.1.

is shown in bold. It can be observed that neither NE nor EW-NE is a good representation strategy for extracting driver-level vectors. Another interesting finding is that K-means is more robust since it always produces the same scores for the Mean, Median and Mixed representations. Moreover, whether EW is used does not affect performance. The performance of Spectral is also not affected by the weight, but it is still the worst of all methods.

Additionally, adding weight to the Mean driver vector creates a negative effect for the GMM algorithm but has no effect for the BGM, which indicates that the Mean vector itself is a comparatively good representation that requires no extra adjustment. The clustering performances of GMM and BGM using the Median vector are worse than

that of the Mean vector, but the EW-Median significantly improves the score for all the indexes. The two models ultimately achieve the same scores using EW-Median. In addition, the GMM fit on EW-Median outperforms the GMM using the Mean in AR, AMI, Hom and FM scores, and BGM using EW-Median achieves the same results as the BGM using the Mean. Hence, the Median with EW adjustment is also a preferable representation choice. For the GMM using eight types of inputs, the Mixed vector performs the best, except that it has lower scores in AR (0.2506), AMI (0.2755) and Hom (0.3065) than those (0.2739, 0.2862, 0.3167) of the EW-Mixed input. The BGM using EW-Mixed has the same scores as EW-Median, and they are slightly better than the BGM fit on Mixed input with respect to the AR, Com and FM scores. Therefore,

Table 11
Driver style information (0: Aggressive, 1: Normal, 2: Calm) with score and fuel consumption statistics.

vid	num_trips	mean_score	std_score	50%_score	mean_fuel	std_fuel	50%_fuel	Style
81	428	22.34861346	13.84575385	20.15603992	40.765404	10.79040752	38.76412538	0
82	484	23.57952904	15.33259986	20.30295133	41.00543212	9.949907576	38.1057007	0
36	551	31.50187844	16.29381575	27.70971787	35.02662375	8.435804682	33.64179011	0
75	493	31.99311076	18.39897031	28.39386961	37.79364104	9.264698923	34.87104849	0
38	277	36.91133587	18.25906538	30.64358534	30.48618927	8.084806157	28.97351724	0
37	430	39.21863392	14.75982025	37.88088576	31.74392113	8.877521984	29.75250719	0
86	389	40.98884084	16.7253042	40.49887578	40.9086891	12.1613908	37.2782772	0
91	310	41.21055698	18.17433985	42.31438414	34.97334723	8.921010986	33.13857399	0
90	605	44.93442837	17.95265894	45.25470673	32.51557135	8.762305004	30.46133904	0
65	696	45.21473413	16.40657928	45.92079218	31.34709337	8.331505123	30.45203168	0
83	709	46.55803018	13.49782408	46.58073938	31.51442944	8.821671959	30.27223045	1
88	191	44.83434863	18.30371169	47.37357256	33.96285592	7.8852015	33.36415811	0
72	479	46.36180852	16.41040272	47.37388843	31.43524013	9.935357268	28.95902015	1
70	548	46.24769071	17.69425756	48.3503265	33.33696608	10.92074378	31.448427	0
55	745	47.49178869	18.81345469	48.50364989	29.58239634	7.715871473	27.85362069	1
47	644	47.88969477	15.58582355	48.99496184	30.85383175	8.270729277	28.34328736	1
60	531	48.19628312	17.32553354	49.1160884	31.67463945	7.765852438	29.81849636	1
58	688	49.2725219	17.33226452	49.75726281	31.85315529	7.781631077	31.81482835	1
46	823	49.68346229	15.03547811	50.00488727	30.71194919	9.078728167	28.67786986	1
66	788	48.85587118	17.43101035	50.42893163	31.52856461	8.197063175	30.47933751	1
43	894	49.69206386	15.82360171	50.97634307	30.65415998	8.942719922	29.83691588	1
69	828	49.44481485	18.325358	51.60988203	30.7102533	8.932942667	30.39374591	1
68	869	49.99180394	17.55845376	51.97751022	30.97247663	9.489826973	29.1368	1
79	499	50.92839389	17.05487076	52.30621573	32.43406272	9.100016569	32.05515914	1
42	894	50.84626219	17.00329938	52.46441649	32.01469608	9.733914525	30.96888735	1
89	564	51.07318959	15.95841481	52.90779177	30.71038311	9.391113598	29.43897952	2
49	841	52.00749867	16.37581335	53.0309681	31.1601027	9.167105201	30.48297193	1
74	844	51.64237452	15.83039928	53.11016899	29.52500418	8.600805597	28.40584731	2
54	738	51.78858436	15.86105126	53.27522061	29.84411801	8.216367559	29.75173258	2
80	577	50.70011159	18.08257565	53.40565689	31.62059336	8.551924936	31.06666543	1
48	783	52.53754241	15.60967846	53.97721391	32.25279433	9.108627394	32.09869632	1
45	995	52.37242485	16.81133137	54.06029522	31.72672052	8.874456075	31.04903001	1
51	404	51.19460173	19.11728761	54.58626875	31.46632304	8.491266153	30.33553365	1
67	937	52.45261232	16.91399506	54.62687836	28.66300581	8.018929397	27.35516361	2
84	793	52.84730751	15.72798301	55.04223991	27.83106328	7.885526262	25.44183624	2
59	704	52.58347241	16.32405856	55.07767085	31.758996	7.857340063	31.61787789	1
78	279	52.5824877	16.74145589	55.23013069	31.20249096	8.678701021	28.71027062	2
64	656	53.86871556	16.08596867	55.66107602	29.52573944	8.739903699	28.34332481	2
77	570	53.22823228	15.82826403	56.18430034	30.68024206	9.348557861	29.65241896	2
56	684	53.94513002	17.25919568	56.31087084	30.39322926	7.851845154	30.7860397	2
73	481	54.35098753	15.64572758	56.41858867	33.76562525	9.245299754	32.50978723	1
53	745	53.41048253	16.55005787	56.52444899	29.24394334	7.373658475	27.78733873	2
61	550	53.44156609	17.31796594	56.56675806	31.11301689	8.405601832	30.09942424	1
62	549	55.37833881	16.76472376	56.65338449	30.59515483	9.235644969	29.62935347	2
44	635	54.63154464	16.85710648	56.75870793	30.52277675	8.758762215	30.2892347	2
57	340	55.01229395	16.0350412	57.94019439	30.116963	7.776257436	28.06225613	2
71	565	55.46941633	16.78218721	58.32404377	30.20571476	10.5502802	29.80092842	2
63	538	56.29364346	15.88584677	58.89769015	29.78100202	7.678623929	29.30130953	2
76	677	55.52781801	16.7816885	59.18172629	28.90561461	7.81986369	26.873	2
87	516	56.69239875	16.54333411	59.89313747	29.75002299	8.627367695	28.07557687	2
50	788	57.12084764	16.47064733	60.16302161	29.76513858	8.358766857	28.76326314	2
85	147	59.63070356	17.64427569	62.66820761	30.79711119	10.75052543	28.65659241	2
52	442	61.46304026	15.16963571	64.12505603	31.43864345	7.998054017	32.31010504	2

the Mixed and EW-Mixed methods can also be considered competing representation methods.

To select the best clustering model, we compare the four models by their highest scores under the five evaluation methods. Fig. 10 shows the comparison bar chart, where Spectral performs the worst of all and K-means is no longer better than GMM and BGM, as it was based on the toy data. BGM outperforms GMM in Hom and AMI by 0.25 and 0.27, respectively, but the FM and Comp scores are lower by 0.46 and 1.93, respectively. Therefore, we choose GMM as the optimal clustering model.

To decide the optimal driver style representation method, we take the best score of each evaluation index that each type of input achieved from the four clustering models for comparison. Table 10 lists the best validation results (%) for eight types of inputs. As can be observed from the values in bold, the Mean and EW-Mixed methods are equally best with respect to the AR score. The Mixed input is close but yields the best performance in terms of other validation methods. Since GMM is taken as the optimal clustering model, either the Mixed or EW-Mixed input is acceptable.

In this section, our proposed trip- and driver-level performance evaluation models are validated with synthetic and real data, respectively. Since there are no available scored trips, we mainly validate our scorecard model with qualitative analysis and business feedback. Next, the experiments on driver style clustering, on the one hand, help us to choose the optimal clustering model and input; on the other hand, they prove the advantage of our present weighting and representation strategies.

6.4. Discussion

In this paper, we first proposed an unsupervised scoring method to rank trip-level driving performance. Since drivers are driving in an uncontrolled environment, their performances vary trip by trip, although they may have a primary driving style. This method aims to quantify these driving performance differences. The method is qualitatively validated by visualization and business feedback. As noted in the experimental section, this method produces a well-formed normal

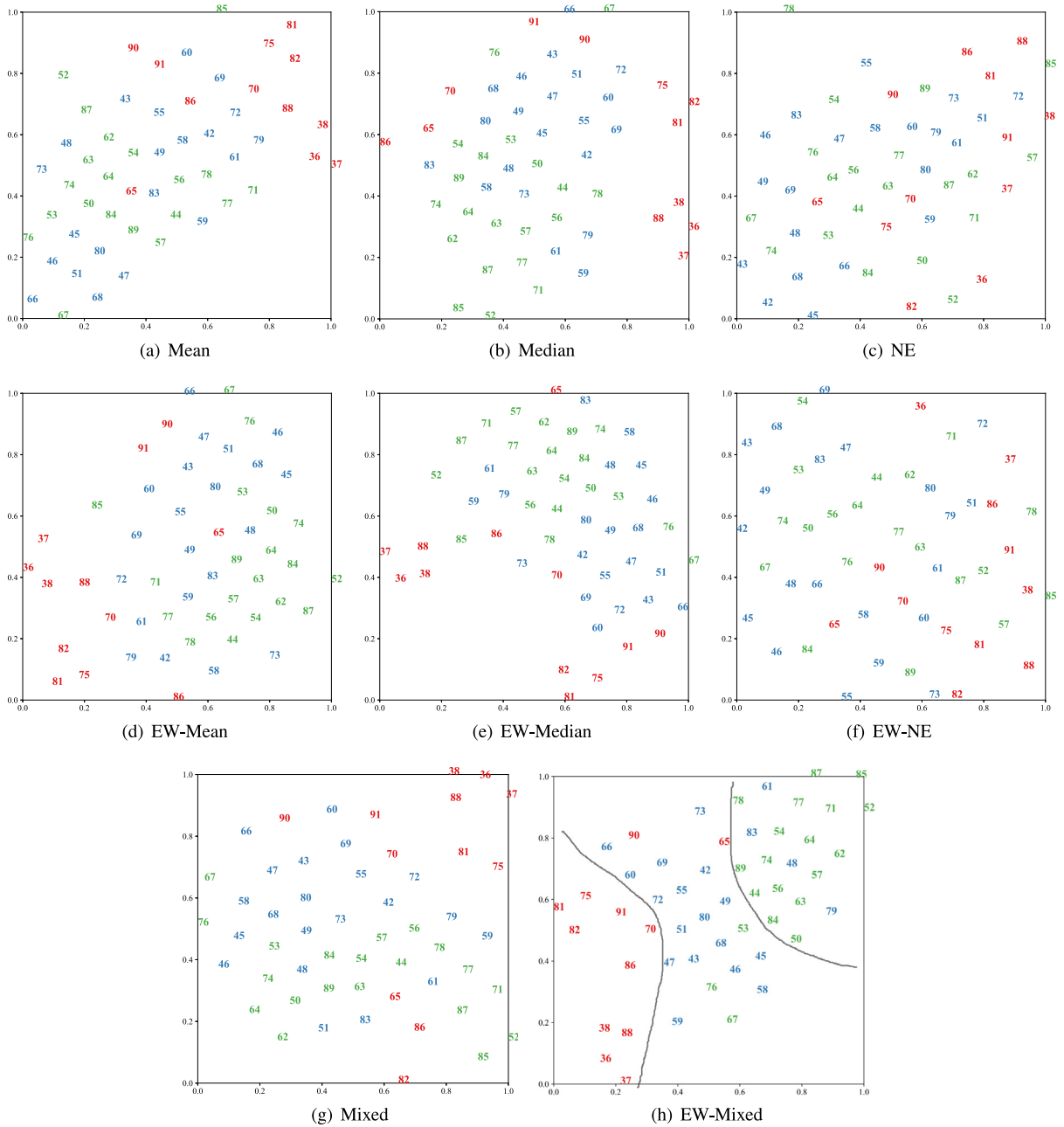


Fig. 12. The low-dimensional plots of driver vectors learned by different representation methods via t-SNE. The driver IDs in red, blue and green denote the aggressive, normal and calm styles, respectively. In this way, it can be intuitively observed which methods can effectively distinguish drivers with different styles. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

distribution based on toy data but is biased in the case of real driving data. This is reasonable in many cases due to noise or insufficiency factors. It can be argued that the scorecard model is a simple and naive approach that lacks technical novelty, but it truly assists in making the model easily explainable.

The driver clustering validation experiments reveal that obtaining the Mean and Mixed rather than the Median vectors as the driver-level representations enables better performance. Furthermore, in the [Appendix](#) section, we draw the t-SNE of all types of clustering inputs, shown in [Fig. 12](#), for an intuitive comparison. As can be observed, the aggressive drivers are easily identified for the Mean, Mixed and EW-Mixed inputs. The other two groups of drivers are difficult to differentiate, but EW-Mixed seems to present a better division by placing the aggressive drivers (red numbers) and calm drivers (green numbers) on two opposite sides and leaving the normal drivers in the middle.

Considering the experimental results in the validation section, it can be concluded that the **EW-Mixed** method incorporates the advantages of the Mean method and the EW-Median method. It could be inferred that when the feature strictly follows a predefined distribution, weighted K-means is the best choice.

7. Conclusion

Driving performance evaluation is essential in fleet management, with benefits in reducing possible traffic violations and accidents, lowering maintenance costs and promoting future self-driving trucks. The real-time requirement of a trip evaluation is a critical business requirement because a fleet manager needs timely information regarding a driver's performance to implement timely intervention measures.

We propose an unsupervised framework that combines the feature distribution mixture and the latent style mixture and contributes to

building a quantitative model for scoring driving performance at the trip level and a qualitative model for classifying driving styles at the driver level. The scorecard model is supported by theoretical evidence and helps produce interpretable scores at the trip level. Hence, it is easy for drivers to understand and managers to accept. The style classification model helps assign an overall label to each driver by aggregating the trip-dependent performance. The trip-level scores are employed as clustering input to consistently link the two models. Next, we suggest an EW strategy considering the importance of each feature and a Mixed driver vectorization method, both of which help improve the modeling process.

This framework is tested on toy data and a real-world dataset. The results prove that our framework not only offers a quantitative trip-level evaluation model that follows a normal distribution but also presents improved clustering models for profiling driver styles. Finally, we also implement a big data solution for an industrial application. This system incorporates our unsupervised framework and enables performing driving performance evaluation in real time.

We hope that our work will help future teams with driver rating tasks by providing insights about the percentile of the population in which each driver resides. Future work could include incorporating other factors (e.g., vehicle load, traffic conditions), and new approaches could be evaluated for segmenting drivers. In addition, considering that the distribution of features might be unclear, we attempt to use the uncertainty quantification method instead of arbitrarily specifying a distribution form. Finally, analysts could achieve a better understanding of real driver behavior before transitioning into fully autonomous driving. Therefore, since the field of self-driving trucks is a promising application realm, our research could help to shape safer and more economical trucks to facilitate intelligent fleet management systems.

CRedit authorship contribution statement

Lin Lu: Data curation, Methodology, Writing - original draft. **Shengwu Xiong:** Conceptualization, Writing - review & editing, Supervision. **Yaxiong Chen:** Proofreading, Writing - review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

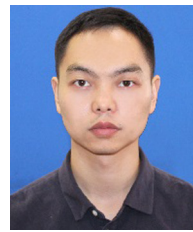
See Figs. 11, 12 and Table 11.

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